

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Industry Problem Solving Task

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO4: *Analyze big data characteristics and challenges across domains including security and application aspects.* (BL4)

Dave's Taxonomy: Articulation (Level 4)
Question Count: 5

Total Marks: 40

Duration :60 Minutes

Code of Conduct

I P Dinesh Reddy (192525399) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____

Assessment Tasks

Industry Problem Solving Task

1. **Retail Data Optimization Problem** - A retail chain operates both physical stores and an online platform. It generates large volumes of sales, inventory, and customer behavior data. The company wants real-time inventory tracking and demand forecasting. It also aims to deliver personalized marketing recommendations to customers. Design a big data architecture to support these requirements. Explain the tools, technologies, and analytics models you would use.

Retail Data Optimization Using Big Data

1. Proposed Big Data Architecture

Data Sources → Data Ingestion → Data Storage → Data Processing → Analytics/ML → Business Applications

- **Data Sources:** Point-of-sale systems, e-commerce transactions, inventory systems, customer clicks, searches, and purchases.
- **Data Ingestion:** **Apache Kafka** collects real-time data streams from stores and the online platform.
- **Data Storage:** **Cloud Data Lake (Amazon S3/Azure Data Lake)** stores structured and unstructured data at large scale.
- **Data Processing:** **Apache Spark** performs batch and real-time processing.
- **Analytics:** Historical and streaming data are analyzed for inventory and customer insights.
- **Applications:** Dashboards, inventory alerts, demand forecasts, and personalized recommendations.

2. Technologies and Tools

Requirement	Technology
Data ingestion	Apache Kafka
Large-scale storage	Cloud Data Lake

Requirement	Technology
Data processing	Apache Spark
Data analytics	Spark SQL / BI Dashboard
Machine Learning	Spark MLlib
Real-time monitoring	Kafka + Spark Streaming

3. Analytics Models

- **Demand Forecasting:** Time-series/ML models predict future product demand using historical sales.
- **Inventory Optimization:** Predictive analytics identifies stock shortages and excess inventory.
- **Recommendation System:** Collaborative filtering recommends products based on customer purchase and browsing behavior.
- **Customer Analytics:** Clustering groups customers according to purchasing patterns.

4. Working

1. Sales and customer data are continuously generated.
2. Kafka collects the incoming real-time data.
3. Data is stored in the cloud data lake.
4. Spark processes both streaming and historical data.
5. ML models forecast demand and generate recommendations.
6. Results are displayed through dashboards and sent to inventory and marketing systems.

5. Security and Scalability

- Use **encryption, authentication, role-based access control, and data masking** to protect customer information.
- Cloud storage and distributed processing provide **horizontal scalability** as data volume increases.
- Backup and replication improve **data availability and reliability**.

6. Expected Outcome

The proposed architecture enables **real-time inventory tracking, accurate demand forecasting, reduced stock-outs, optimized inventory, and personalized customer marketing** while supporting large-scale retail data processing.

2. **Social Media Fraud Detection Problem** - A social media platform processes billions of user interactions daily. These include posts, comments, likes, and shares. The platform wants to detect fake accounts and misinformation quickly. Real-time or near real-time processing is required. Design a scalable big data solution for this use case. Explain the role of machine learning and stream processing frameworks.

Social Media Fraud Detection Using Big Data

1. Proposed Big Data Solution

User Data → Kafka → Stream Processing → ML Models → Fraud Detection → Alerts/Dashboard

- **Data Sources:** Posts, comments, likes, shares, account activity, and user profiles.
- **Data Ingestion:** **Apache Kafka** collects billions of real-time interactions.
- **Storage:** A **cloud data lake** stores historical and streaming data.
- **Stream Processing:** **Apache Spark Streaming** processes incoming data with low latency.
- **Machine Learning:** ML models identify fake accounts and misinformation.
- **Output:** Suspicious accounts/content are flagged and sent for review or automatic action.

2. Role of Machine Learning

- **Classification:** Classifies accounts as genuine or suspicious.
- **Anomaly Detection:** Identifies unusual patterns such as abnormal posting or sharing behavior.
- **NLP:** Analyzes text to identify potentially misleading or harmful information.

- **Graph Analysis:** Examines relationships between accounts to detect coordinated fake-account networks.

3. Stream Processing Framework

Apache Kafka provides high-throughput event ingestion, while **Spark Streaming** performs near-real-time processing and analysis.

4. Working

1. User interactions continuously enter Kafka.
2. Kafka distributes data to processing systems.
3. Spark Streaming analyzes events in real time.
4. ML models calculate fraud or misinformation risk.
5. High-risk activities are flagged.
6. Alerts are generated for moderators or automated actions.
7. Processed data is stored for historical analysis and model improvement.

5. Security and Scalability

- Apply **encryption, authentication, access control, and privacy protection**.
- Use distributed Kafka and Spark clusters to handle billions of events.
- Replication and fault tolerance ensure continuous service.

6. Expected Outcome

The solution provides **fast and scalable detection of fake accounts and misinformation**, reduces fraudulent activity, and supports near-real-time monitoring of massive social-media data.

3. **Government Data Platform Problem** A government agency is building a national data platform. It combines census, economic, and geospatial datasets. Different departments need access to shared and integrated data. The system must ensure interoperability across multiple agencies. Strict privacy and security requirements must be followed. Propose a governance and data management strategy.

Government Data Platform – Governance and Data Management Strategy

1. Proposed Architecture

Government Departments → Data Ingestion → Central Data Lake → Data Integration →

Governance & Security → Shared Data Platform

- **Data Sources:** Census, economic, and geospatial databases.
- **Data Ingestion:** APIs and ETL pipelines collect data from different departments.
- **Data Storage:** A centralized **cloud data lake** stores integrated datasets.
- **Data Processing:** **Apache Spark** processes and transforms large datasets.
- **Data Access:** Departments access approved datasets through APIs and dashboards.

2. Data Governance Strategy

- Establish common **data standards and formats** across departments.
- Maintain a **data catalog** containing dataset descriptions and ownership.
- Define **data ownership and access policies** for each department.
- Implement **data quality checks** to ensure accuracy and consistency.
- Maintain metadata for easy discovery and interoperability.

3. Interoperability

- Use standardized data formats and APIs.
- Establish common schemas and identifiers.
- Use an API-based integration layer for secure data exchange.
- Apply consistent metadata standards across agencies.

4. Security and Privacy

- Use **encryption** for data at rest and in transit.
- Apply **role-based access control (RBAC)**.
- Use authentication and authorization for all users and applications.
- Apply **data masking/anonymization** to sensitive information.
- Maintain audit logs to monitor data access and activities.

5. Data Management Process

1. Collect data from different government departments.
2. Validate and clean the incoming data.
3. Transform data into common formats.
4. Store it in the centralized data platform.
5. Apply governance and security policies.
6. Provide authorized departments with shared data access.
7. Continuously monitor data quality and security.

6. Expected Outcome

The proposed strategy provides **integrated, interoperable, secure, and governed government data**, enabling departments to share information efficiently while protecting privacy and maintaining data quality.

4. Banking Fraud Detection Problem

A bank processes millions of financial transactions every minute. It needs to detect fraudulent activities in real time. The system must analyze historical and streaming transaction data.

Low latency and high accuracy are critical requirements. Design a big data pipeline for fraud detection. Explain the technologies and algorithms you would implement.

Banking Fraud Detection Using Big Data

1. Proposed Big Data Pipeline

Transactions → Kafka → Stream Processing → Feature Engineering → ML Model → Fraud Score → Alert/Action

- **Data Sources:** Credit/debit card transactions, online banking transactions, customer profiles, and historical transaction records.
- **Data Ingestion:** **Apache Kafka** receives transactions in real time.
- **Storage:** A **cloud data lake** stores historical transaction data.
- **Processing:** **Apache Spark Streaming** processes transactions with low latency.
- **Analytics:** Machine learning models identify suspicious transactions.
- **Output:** Fraudulent transactions generate immediate alerts or are blocked.

2. Technologies Used

Requirement	Technology
Real-time ingestion	Apache Kafka
Stream processing	Apache Spark Streaming
Data storage	Cloud Data Lake
Batch processing	Apache Spark
Machine Learning	Spark MLlib
Monitoring	Real-time Dashboard

3. Fraud Detection Algorithms

- **Random Forest:** Classifies transactions as genuine or fraudulent.

- **Logistic Regression:** Calculates the probability of fraud.
- **Anomaly Detection:** Identifies unusual transaction patterns.
- **Rule-Based Detection:** Detects predefined suspicious conditions.

4. Working

1. Transaction data enters the system through Kafka.
2. Spark Streaming processes transactions immediately.
3. Customer and historical data are used to generate relevant features.
4. ML models calculate a fraud probability/risk score.
5. High-risk transactions trigger alerts or blocking.
6. Transaction results are stored for future analysis and model training.

5. Security and Reliability

- Use **encryption** for transaction data.
- Apply **authentication and role-based access control**.
- Maintain audit logs for transaction monitoring.
- Use data replication and fault-tolerant distributed processing.
- Protect sensitive customer information through masking and secure access.

6. Expected Outcome

The pipeline enables **real-time, low-latency and scalable fraud detection**, improving transaction security while reducing false positives and financial losses.

5. Agriculture Smart Farming Problem

A smart farming system collects soil, weather, and crop data. Sensors and drones provide real-time field information. Farmers want insights to improve yield and resource usage. Data is generated from multiple distributed sources. Design a big data solution for precision agriculture. Explain how analytics can support decision-making.

Smart Farming Using Big Data Analytics

1. Proposed Big Data Architecture

Sensors & Drones → IoT Gateway → Kafka → Cloud Data Lake → Spark Analytics → ML Models → Farmer Dashboard

- **Data Sources:** Soil sensors, weather stations, drones, satellite images, and crop monitoring systems.
- **Data Ingestion:** **Apache Kafka** collects real-time sensor and field data.
- **Storage:** A **cloud data lake** stores structured and unstructured agricultural data.
- **Processing:** **Apache Spark** processes large-scale historical and streaming data.
- **Analytics:** Machine learning models analyze crop, soil, and weather conditions.
- **Output:** Insights are provided through dashboards and alerts.

2. Analytics Applications

- **Crop Yield Prediction:** Predicts expected crop production using historical and current data.
- **Irrigation Optimization:** Determines when and how much water is required.
- **Fertilizer Optimization:** Recommends fertilizer usage based on soil conditions.
- **Crop Disease Detection:** Drone images and ML models identify possible crop diseases.
- **Weather Analysis:** Predicts weather conditions that may affect crops.

3. Working

1. Sensors and drones continuously collect field data.
2. IoT gateways transmit the data to Kafka.
3. Data is stored in the cloud data lake.

4. Spark processes and analyzes the data.
5. ML models generate predictions and recommendations.
6. Farmers receive insights through dashboards or alerts.
7. Farmers use these insights to make timely decisions.

4. Technologies Used

Requirement	Technology
Data collection	IoT Sensors & Drones
Data ingestion	Apache Kafka
Storage	Cloud Data Lake
Processing	Apache Spark
Machine Learning	Spark MLlib
Visualization	BI/Farmer Dashboard

5. Security and Scalability

- Use **authentication, encryption, and access control** to protect agricultural data.
- Cloud storage provides scalable capacity for growing datasets.
- Distributed processing enables analysis of data from multiple fields and sources.
- Data backup and replication improve reliability.

6. Expected Outcome

The solution supports **precision agriculture**, improves crop yield, reduces water and fertilizer wastage, enables early disease detection, and helps farmers make **data-driven decisions**.

Industry Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Industry Context	20%	Demonstrates strong understanding of real-world problem context	Good understanding	Basic understanding	Weak understanding
Application of Big Data Concepts	25%	Applies highly relevant big data ideas and tools	Mostly appropriate application	Partial application	Weak application
Solution Design	25%	Solution is practical, scalable, and well-reasoned	Reasonable solution	Basic solution with gaps	Unclear solution
Security / Risk Awareness	15%	Strong awareness of security and operational risks	Reasonable awareness	Limited awareness	Poor awareness
Clarity and Professionalism	15%	Well-structured and professional response	Generally clear	Moderately clear	Poorly presented

Total Marks Awarded: